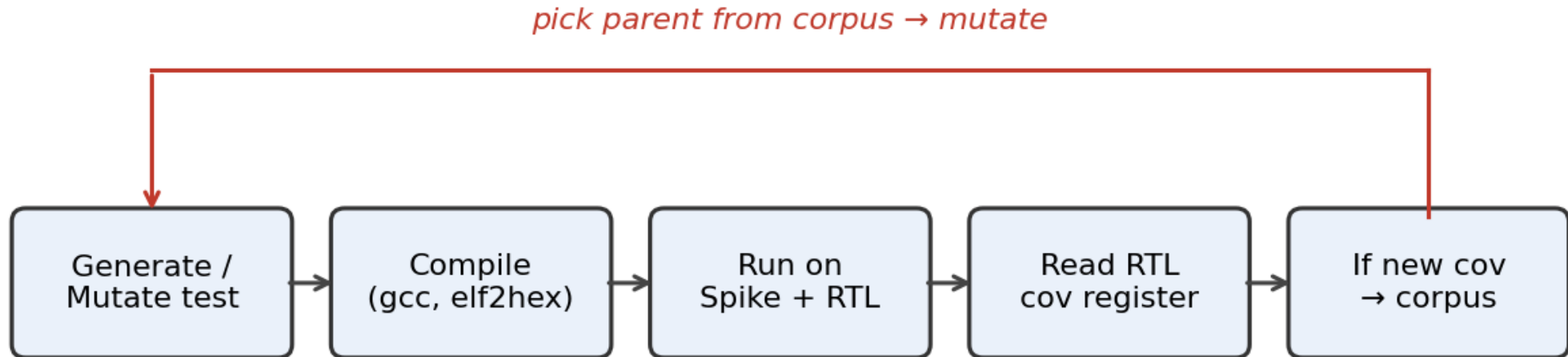


Tool Demo: Running the Modified DifuzzRTL

What The Demo Proves

- 1 We can launch the modified DifuzzRTL experiment from one command.
- 2 The job is submitted to the cluster, so it runs the same way as the real campaign.
- 3 The output folders contain coverage logs that can be checked later.

One Picture of the Tool



DifuzzRTL repeatedly generates tests, runs them on the hardware model, and records coverage.

Files to point at before running

Fuzzer/src/mutator.py

Main fuzzer logic changed for this project

Fuzzer/src/mutator_old.py

Original artifact version kept for reference

Fuzzer/Fuzzer.py

Connects the fuzzer loop to the updated logic

scripts/submit_sweep_by_seed.sh

Script to submit the jobs

server_runs/

Where output folders and coverage logs are written

Command 1: Enter the Artifact Repo

Start from the submitted artifact directory.

```
cd difuzz-rtl-sim/difuzz-rtl-sim
```

What to notice: all later commands are run from this repo root.

Command 2: Important Files

```
ls Fuzzer/src/mutator.py Fuzzer/src/mutator_old.py \
Fuzzer/Fuzzer.py scripts/submit_sweep_by_seed.sh server_runs
```

What to notice: there are only a few files to point at before launching the tool.

Command 3: Helper Script

The helper script is the normal way to launch the demo jobs.

```
sed -n '1,16p' scripts/submit_sweep_by_seed.sh
```

What to notice: the script takes one seed number and submits the experiment jobs.

Command 4: Submit One Seed

```
./scripts/submit_sweep_by_seed.sh 1
```

What to notice: the script submits jobs and prints the queue command to check them.

Command 5: The cluster queue

This shows the tool run was accepted by the cluster queue.

```
squeue -u $USER
```

What to notice: the job names appear in the queue with running or pending status.

After jobs finish: How to Find Outputs

Use this only if completed output folders are already present.

```
ls server_runs/sweep/seed_1
```

What to notice: each subfolder is one run and contains a coverage log.

How to see the coverage log

```
$ tail -3 server_runs/sweep/seed_1/boom-evict/cov_log_*.txt
```

```
43273.45    12376    93379
43274.62    12378    93381
43276.48    12381    93383
```

elapsed seconds

time since the run started

iteration

test number

coverage

hardware behavior reached

Live demo commands

```
cd difuzz-rtl-sim/difuzz-rtl-sim
./scripts/submit_sweep_by_seed.sh 1
queue -u $USER
```

After the queue appears, explain that long runs finish later and write coverage logs under `server_runs/sweep/seed_1`.